

On the learning representation of LLM : A concept-oriented study

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On deep learning intuition

On Mechanistic Interpretability

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Classic Machine Learning

$$\min_{\theta \in \Theta} \mathcal{C}(\theta) := \mathbb{E}_{(x,y) \sim \mathcal{D}} [\mathcal{C}(f(\theta; x), y)] \quad (\text{ML})$$

- θ : model parameters, Θ : parameter space
- $f(\theta; x)$: model function, $\mathcal{C}$: cost function
- $\mathcal{D}$: data distribution, (x, y) : input-target pairs

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SGD Discrete Update:

$$\theta_{t+1} = \theta_t - \eta \cdot \mathbb{E}_{(x,y) \sim \mathcal{B}} (\nabla_{\theta} \mathcal{C}(f(\theta_t; x), y)) \quad (1)$$

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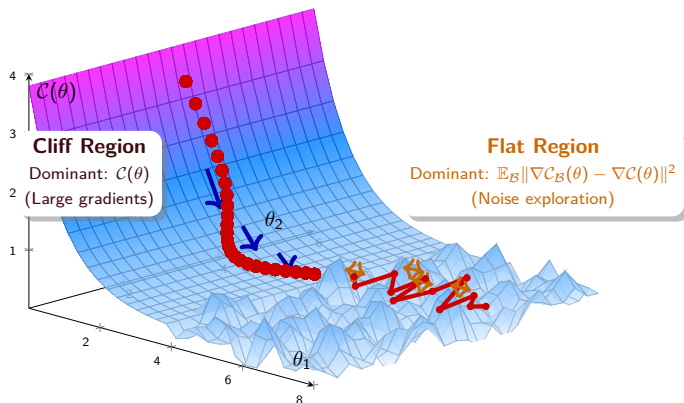
SGD Implicit Regularization [1]:

$$\mathcal{C}_{\text{SGD}}(\theta) = \mathcal{C}(\theta) + \frac{\eta}{4} \|\nabla \mathcal{C}(\theta)\|^2 + \frac{\eta}{4} \mathbb{E}_{\mathcal{B}} \|\nabla \mathcal{C}_{\mathcal{B}}(\theta) - \nabla \mathcal{C}(\theta)\|^2 \quad (2)$$

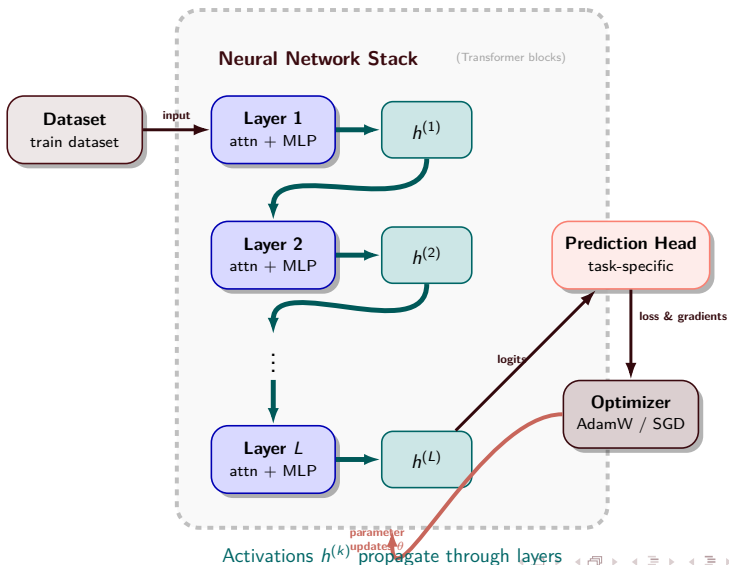
Where $\mathcal{C}_{\mathcal{B}}(\theta)$ is the cost on the mini-batch $\mathcal{B}$.

SGD Dynamics: From Cliff to Flat Landscape

SGD Trajectory: *Smooth* → *Erratic*

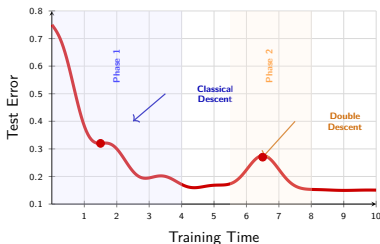


Deep learning - pipeline



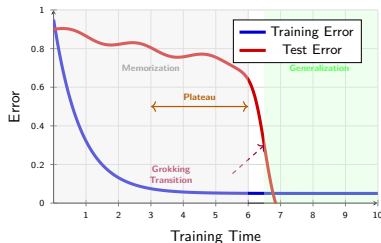
The Big Question: Why does deep learning generalize so well?

Double Descent Phenomenon [2]



Test error initially decreases, then increases, then decreases again

Grokking Phenomenon [3]



Test error plateaus during memorization, then suddenly drops during generalization

Intuition: Both phenomena challenge classical learning theory. **Double descent** shows that larger models can generalize better even when appearing to overfit, while **grokking** demonstrates that generalization can emerge suddenly after prolonged memorization phases.

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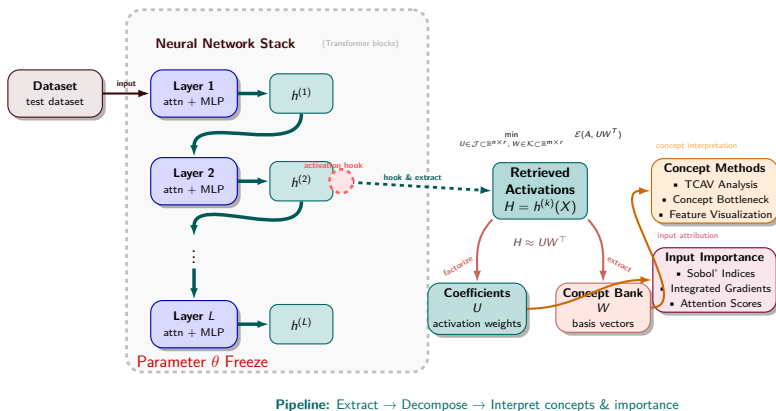
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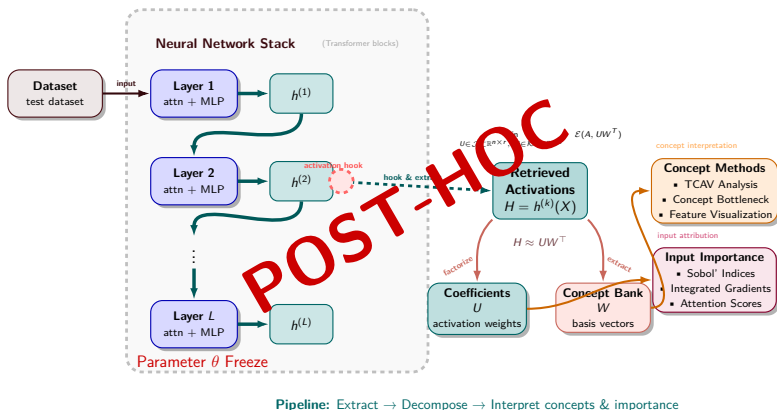
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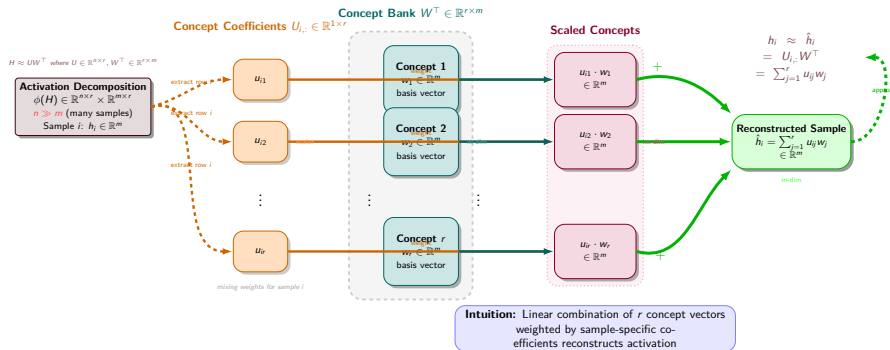
Concept-Oriented Mechanistic Interpretability



Concept-Oriented Mechanistic Interpretability



Concept-Based Sample Reconstruction



Unsupervised Concept Extractor Methods

General Optimization Framework:

$$\phi(h_\theta(x)) \in \arg \min_{W \in \mathcal{K}} \left(\min_{U \in \mathcal{J}} \mathcal{E}(h_\theta(x), UW^\top) \right)$$

Method	$\mathcal{E}(A, B)$	$\mathcal{J}$ (Encoded Activations)	$\mathcal{K}$ (Concept Dictionary)
NMF Family			
NMF (vanilla)	$\ A - B\ _F^2$	$\mathbb{R}_+^{n \times r}$	$\mathbb{R}_+^{m \times r}$
Convex NMF	$\ A - B\ _F^2$	$\{U = AH, H \geq 0, \mathbf{1}_m^\top H = \mathbf{1}_r^\top\}$	$\mathbb{R}_+^{m \times r}$
Semi-NMF	$\ A - B\ _F^2$	$\mathbb{R}^{n \times r}$	$\mathbb{R}_+^{m \times r}$
Statistic / SVD-ICA Family			
SVD	$\ A - B\ _F^2$	$\{U^\top U = \Sigma(A)\}$	$\{V^\top V = \Sigma(A)\}$
PCA	$\ A - B\ _F^2$	$\{U : \mathbf{1}_n^\top U = 0\}$	$\{W^\top W = I_r\}$
Sparse PCA	$\ A - B\ _F^2$	$\{U : \mathbf{1}_n^\top U = 0, \ U\ _1 \leq \tau\}$	$\{W^\top W = I_r\}$
ICA	$\ A - B\ _F^2$	$\mathcal{U}_{\text{ICA}}(A, \tau)$	$\mathbb{R}^{m \times r}$
Sparse / Dictionary Family			
Dictionary Learning	$\ A - B\ _F^2$	$\{U : \ U\ _1 \leq \tau\}$	$\mathbb{R}^{m \times r}$
Sparse Auto-Encoder	$\ A - B\ _F^2$	$\{U = E_\theta(A)\}$	$\{W : \ W\ _1 \leq \tau\}$
k-Means	$\ A - B\ _F^2$	$\{U \in \{0, 1\}^{n \times r}, U\mathbf{1}_r = \mathbf{1}_n\}$	$\mathbb{R}^{m \times r}$

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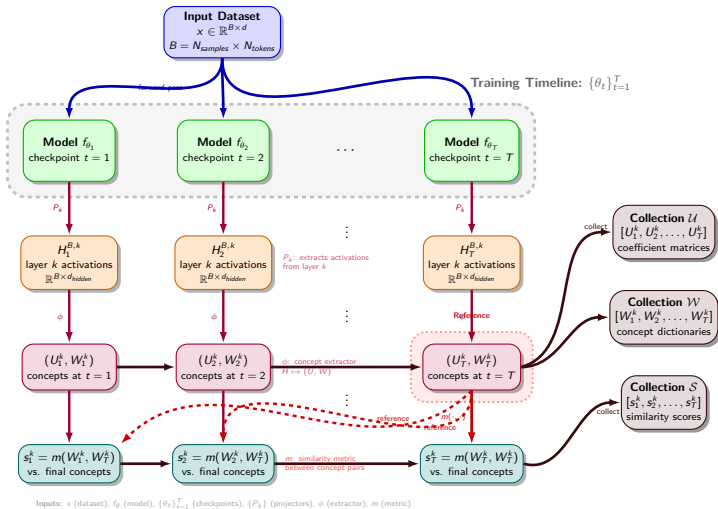
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Algorithm: Concept Evolution Across Training Checkpoints

Algorithm: Compute concept representations across checkpoints and similarity to final-step concepts.



Empirical Learning Dynamics: Setup

Large Language Model:

EuroBERT-210m[4]

MLM Head

masked language modeling

11 Boosted Transformers

self-attention + MLP
210M parameters

Tokenizer

BERT WordPiece

Example Datasets:

RottenTomatoes

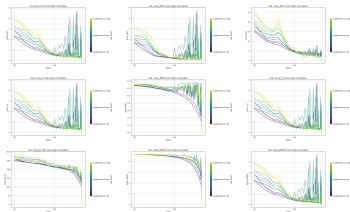
Movie and TV series reviews with critic ratings and sentiment labels.

Wikipedia

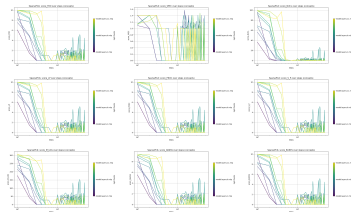
Encyclopedia articles covering diverse topics and knowledge domains.

Setup: Analyze concept dynamics
in EuroBERT during training

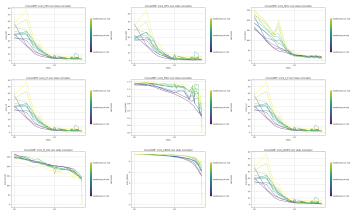
Empirical Learning Dynamics



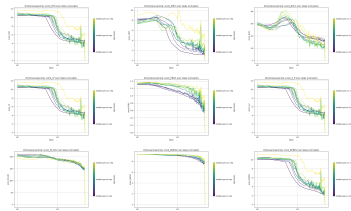
ICA Concepts



Sparse PCA Concepts



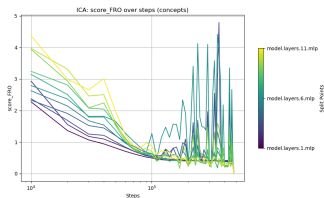
Convex NMF Concepts



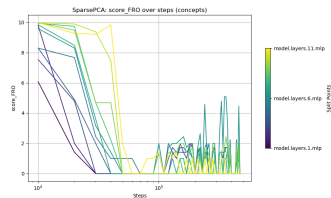
Dictionary Learning Concepts



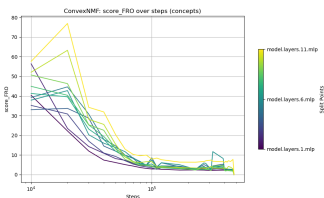
Empirical Learning Dynamics : Focus



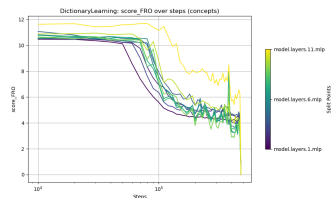
ICA Concepts



Sparse PCA Concepts

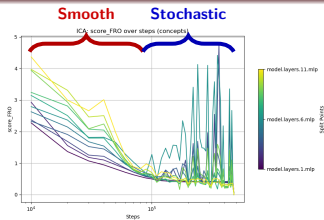


Convex NMF Concepts

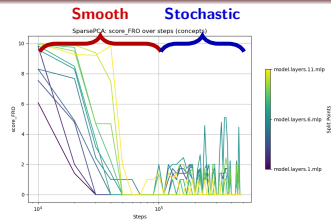


Dictionary Learning Concepts

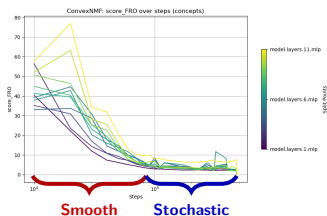
Empirical Learning Dynamics : Focus



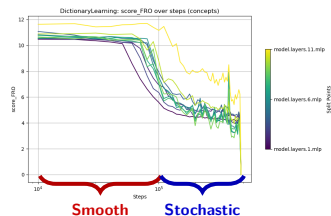
ICA Concepts



Sparse PCA Concepts



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Learning Dynamics : Parameter Convergence - 1

Lipschitz Continuity - A function $f : \Omega \rightarrow \Gamma$ is Lipschitz continuous if and only if there exists $L > 0$ such that, for all $x, y \in \Omega$,

$$\|f(y) - f(x)\|_{\Gamma} \leq L\|y - x\|_{\Omega}.$$

Here, $\|\cdot\|_{\Omega}$ and $\|\cdot\|_{\Gamma}$ denote norms on the spaces Ω and Γ respectively.

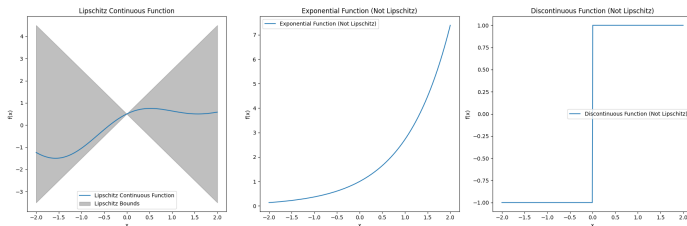


Figure: Lipschitz Visualization (GitHub:Raphael-Bernas).

Learning Dynamics : Parameter Convergence - 2

- **Assumption A:** The input set $\mathcal{X}$ is bounded and finite-dimensional.
- **Assumption B:** The parameter set Θ is compact and finite-dimensional.
- **Assumption C:** The model f_θ is a Transformer-based LLM with activation functions $\{\sigma_i\}_{i=1}^N$ that are locally Lipschitz.

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Extractor Algorithm-Based Estimator: $\hat{\phi} : A \mapsto (\hat{U}(A), \hat{W}(A))$

- (i) The algorithm uses only a fixed *maximum iteration* stopping criterion.
- (ii) Each computational step of the algorithm is locally Lipschitz with respect to its inputs.
- (iii) All stochastic initializations are determined prior to execution.

Learning Dynamics : Parameter Convergence - 3

Theorem: Under Assumptions A, B, C, and the Extractor Algorithm-Based Estimator conditions (i), (ii), (iii), we have that for any layer k and input $x \in \mathcal{X}$, for all training steps t :

$$d(\hat{\phi}(P_k(f_{\theta_t}, x)), \hat{\phi}(P_k(f_{\theta_T}, x))) \leq L_d L_\phi L_f(x) \|\theta_t - \theta_T\|.$$

Which implies,

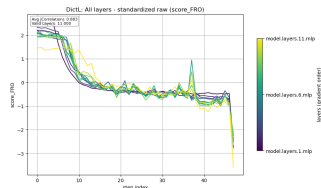
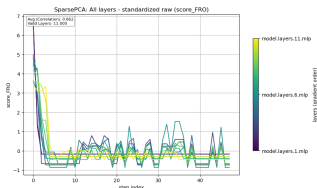
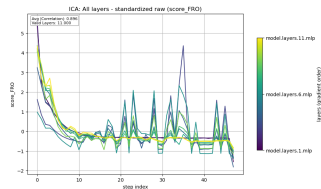
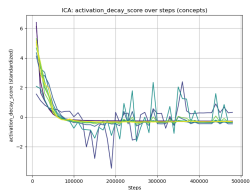
$$s_t^k = O(\|\theta_t - \theta_T\|).$$

This means that the **learning dynamics** of the **concept** representations are directly **controlled** by the **parameter convergence** of the model during training.

Learning Dynamics : Signal Analysis - 1

Activation decay signal for each layer k :

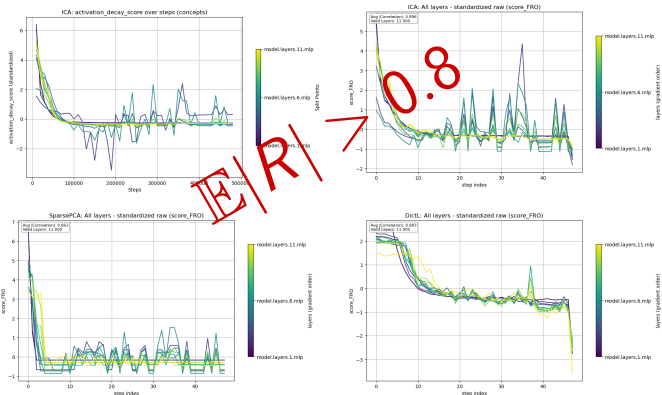
$$a_t^k = \|P_k(f_{\theta_t}, x) - P_k(f_{\theta_T}, x)\|_F = \|H_t^{(k)} - H_T^{(k)}\|_F,$$



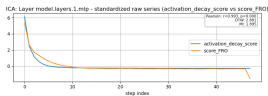
Learning Dynamics : Signal Analysis - 1

Activation decay signal for each layer k :

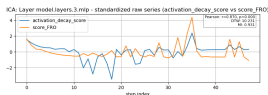
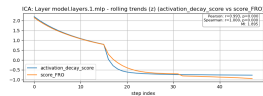
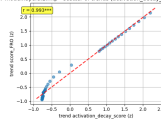
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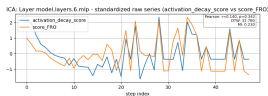
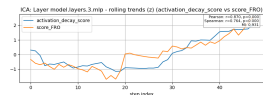
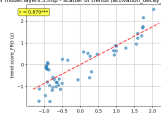
Learning Dynamics : Signal Analysis - 2



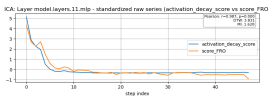
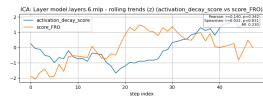
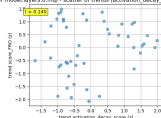
ICA: Layer model.layers.1.mip - scatter of trends (activation_decay_score vs score_FRO)



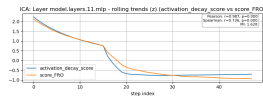
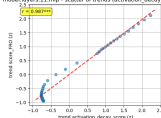
ICA: Layer model.layers.3.mip - scatter of trends (activation_decay_score vs score_FRO)



ICA: Layer model.layers.6.mip - scatter of trends (activation_decay_score vs score_FRO)

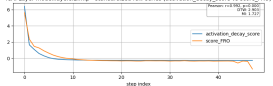


ICA: Layer model.layers.11.mip - scatter of trends (activation_decay_score vs score_FRO)

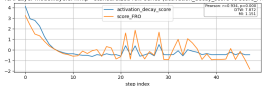


Learning Dynamics : Signal Analysis - 2

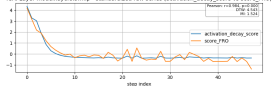
ICA: Layer model.layers.2.mip - standardized raw series (activation_decay_score vs score_FRO)



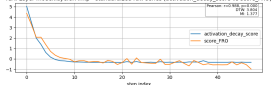
ICA: Layer model.layers.4.mip - standardized raw series (activation_decay_score vs score_FRO)



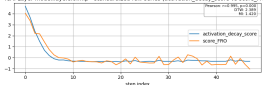
ICA: Layer model.layers.5.mip - standardized raw series (activation_decay_score vs score_FRO)



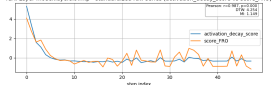
ICA: Layer model.layers.7.mip - standardized raw series (activation_decay_score vs score_FRO)



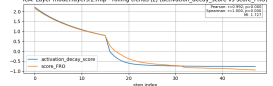
ICA: Layer model.layers.8.mip - standardized raw series (activation_decay_score vs score_FRO)



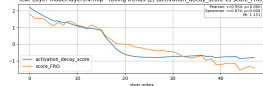
ICA: Layer model.layers.9.mip - standardized raw series (activation_decay_score vs score_FRO)



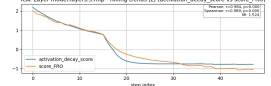
ICA: Layer model.layers.2.mip - rolling trends (z) (activation_decay_score vs score_FRO)



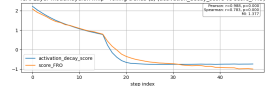
ICA: Layer model.layers.4.mip - rolling trends (z) (activation_decay_score vs score_FRO)



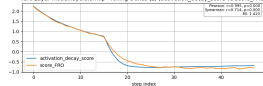
ICA: Layer model.layers.5.mip - rolling trends (z) (activation_decay_score vs score_FRO)



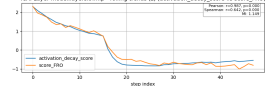
ICA: Layer model.layers.7.mip - rolling trends (z) (activation_decay_score vs score_FRO)



ICA: Layer model.layers.8.mip - rolling trends (z) (activation_decay_score vs score_FRO)



ICA: Layer model.layers.9.mip - rolling trends (z) (activation_decay_score vs score_FRO)



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Giving Meaning to Concepts

The Concepts Manifold

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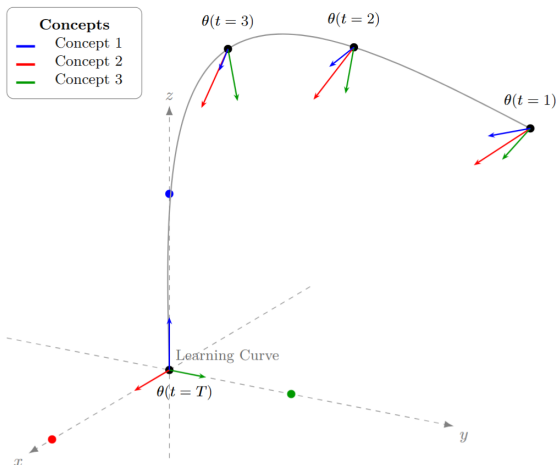
The Concepts Manifold

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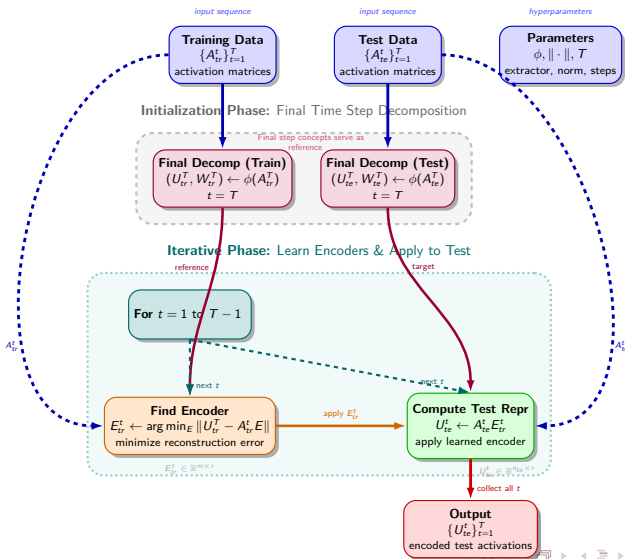
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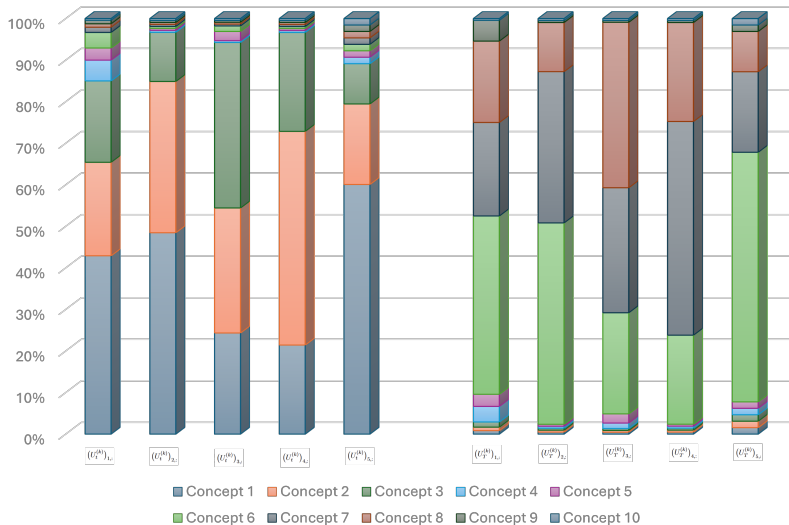
The Impossible Visualization of High-Dimensional Concepts



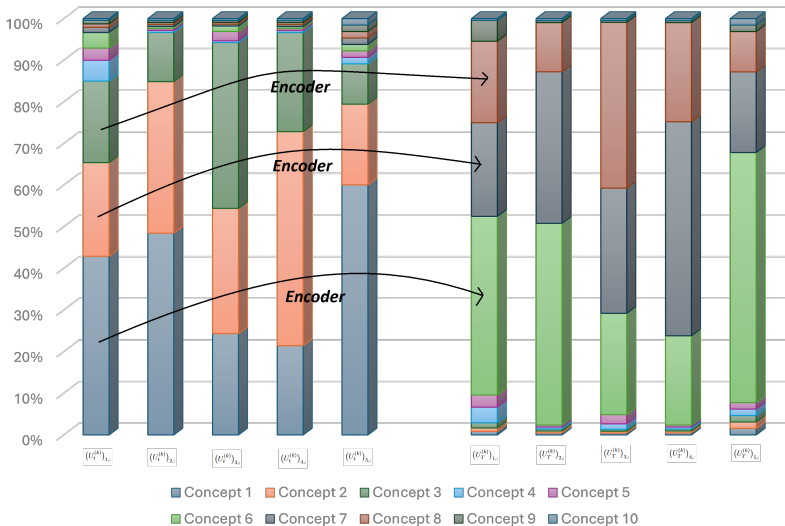
Algorithm: Activation-based Representation Learning Algorithm



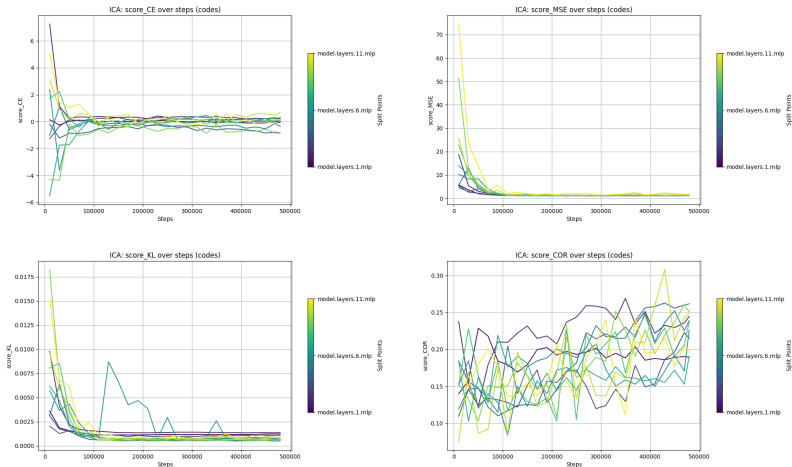
The Concept Distribution Evolution: An Example Histogram



The Concept Distribution Evolution: An Example Histogram



The Concept Distribution Evolution: Learning Dynamics



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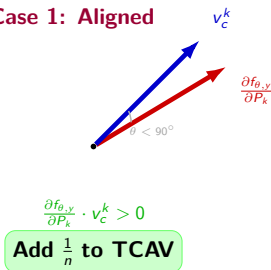
6 References

Testing with CAV: The Method [5]

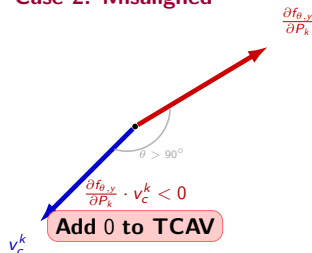
$$\text{TCAV}_{y,c}^k = \mathbb{E}_{x \sim \mathcal{D}_y} \left[\mathbb{I} \left(\frac{\partial f_{\theta,y}(x)}{\partial P_k(f_{\theta}, x)} \cdot v_c^k > 0 \right) \right],$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. It measures the fraction of inputs for which moving in the direction of the concept increases the probability of predicting class y .

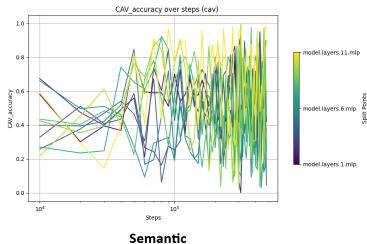
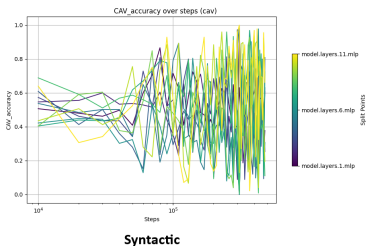
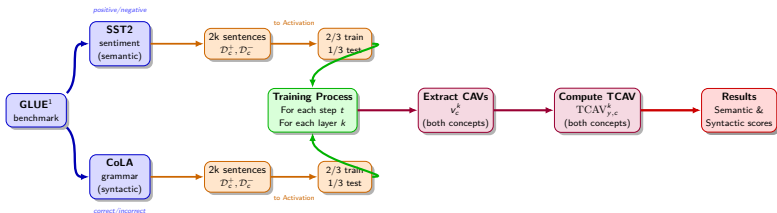
Case 1: Aligned



Case 2: Misaligned



TCAV: Learning Dynamics



TCAV: Regime-wise analysis

Layer	Syntactic			Semantic		
	First	Second	Overall	First	Second	Overall
1	0.597 (0.145)	0.440 (0.115)	0.514 (0.124)	0.571 (0.137)	0.581 (0.109)	0.561 (0.142)
2	0.514 (0.077)	0.594 (0.184)	0.498 (0.184)	0.535 (0.186)	0.635 (0.179)	0.536 (0.195)
3	0.543 (0.146)	0.480 (0.150)	0.473 (0.202)	0.329 (0.128)	0.339 (0.191)	0.455 (0.201)
4	0.385 (0.180)	0.341 (0.237)	0.445 (0.220)	0.423 (0.183)	0.577 (0.294)	0.514 (0.220)
5	0.483 (0.176)	0.506 (0.264)	0.496 (0.224)	0.441 (0.235)	0.367 (0.201)	0.441 (0.205)
6	0.342 (0.116)	0.574 (0.250)	0.443 (0.221)	0.308 (0.126)	0.528 (0.270)	0.427 (0.219)
7	0.606 (0.156)	0.484 (0.221)	0.479 (0.240)	0.650 (0.205)	0.277 (0.232)	0.463 (0.252)
8	0.541 (0.087)	0.337 (0.203)	0.504 (0.228)	0.502 (0.175)	0.541 (0.248)	0.473 (0.232)
9	0.501 (0.168)	0.725 (0.162)	0.514 (0.251)	0.663 (0.173)	0.572 (0.291)	0.541 (0.241)
10	0.512 (0.230)	0.619 (0.247)	0.566 (0.258)	0.684 (0.162)	0.491 (0.284)	0.548 (0.263)
11	0.593 (0.172)	0.657 (0.275)	0.573 (0.284)	0.606 (0.277)	0.678 (0.190)	0.672 (0.241)
$\Sigma\Delta$		+0.142			-0.127	
$\Sigma\omega\Delta$		+0.125			-0.292	

- **Shaded:** Higher regime mean for each layer/modality
- **Blue/Red:** Lower/Higher overall means between overall columns
- $\Sigma\Delta$: Simple sum of (Second - First) regime differences
- $\Sigma\omega\Delta$: Variance-weighted aggregate of regime changes

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Encoding The Concept Distribution

Giving Meaning to Concepts

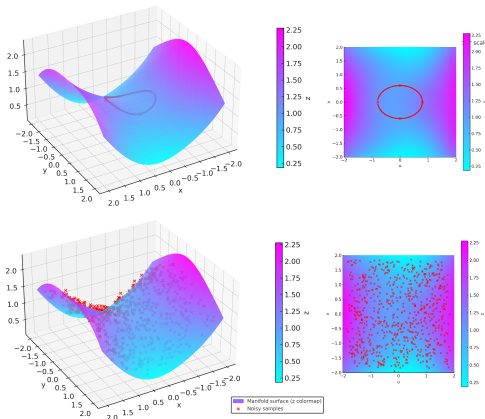
The Concepts Manifold

4 Conclusion

5 Annex

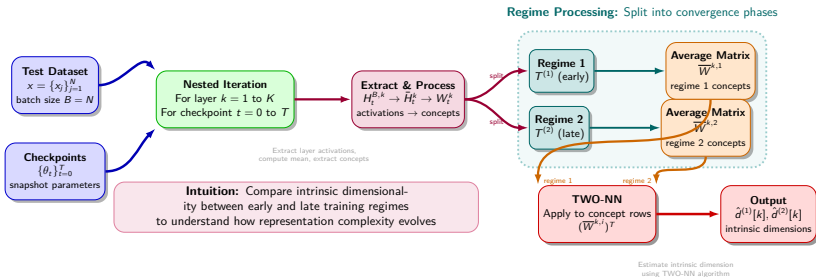
6 References

What is a Manifold ?



Intuition: A manifold is a surface that *locally* looks like Euclidean space. Even though the surface is curved in 3D, any path on it can be described using only **2 parameters** -> 2 degrees of freedom.

The Concepts Manifold: Estimating Intrinsic Dimension

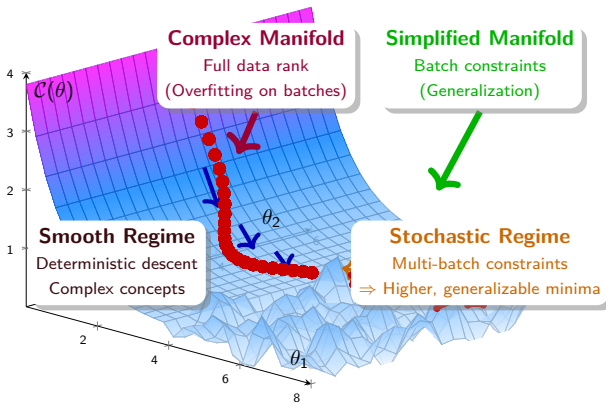


The Concepts Manifold: Results Overview

Layer	Non-sparse methods						Sparse methods			
	SVD		ICA		Semi-NMF		Dict. Learning		K-Means	
	First	Second	First	Second	First	Second	First	Second	First	Second
1	34.91	24.70	46.98	31.09	34.06	24.95	13.17	15.25	11.10	15.57
2	28.72	25.05	37.94	23.79	27.04	27.97	12.35	13.58	9.61	9.98
3	35.84	34.02	23.14	8.55	22.77	11.29	9.76	10.19	3.05	2.52
4	40.35	34.27	17.54	16.51	20.01	12.73	11.20	8.03	3.54	4.01
5	42.18	37.83	22.85	20.86	21.58	18.38	8.67	7.83	3.39	3.91
6	46.52	36.01	14.58	9.99	10.42	8.76	7.28	7.79	3.18	3.13
7	36.96	37.08	29.10	21.27	15.24	24.22	7.33	6.00	6.32	6.30
8	49.08	38.95	20.17	21.99	40.13	22.32	9.04	6.04	8.58	8.67
9	48.40	55.21	20.69	14.67	20.96	13.78	10.69	8.21	8.45	17.47
10	75.07	57.50	44.47	37.67	34.07	32.38	10.64	7.12	7.74	14.02
11	68.98	55.10	20.36	20.79	24.42	23.50	8.30	5.69	4.78	6.70
Mean	46.09	39.61	27.07	20.65	24.61	20.03	9.86	8.70	6.34	8.39

Estimated intrinsic dimension per layer and regime. For each method we report the block-mean intrinsic dimension in the **First** and **Second** regimes. The **gray** cell indicates which regime (first or second) yields the *lower* intrinsic dimension for that layer-method pair. The final row (**Mean**) gives the average intrinsic dimension across layers.

The Concepts Manifold: Interpretation



Satisfying diverse batch gradients forces higher minima that generalize better, leading to manifold dimension reduction through concept simplification

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Conclusion - 1

Main Contributions:

- **Unified ML Framework:** Proposed general mechanistic interpretability framework unifying classical methods
- **Two-Regime Confirmation:** Empirically confirmed dual training phases in LLMs beyond noisy loss curves
- **Concept Distribution Stability:** Showed concept distributions remain stable despite individual vector noise
- **Semantic Validation:** TCAV analysis confirmed regimes exist at human-interpretable concept level even if need more refined concepts
- **Manifold Simplification:** **Key finding:** Intrinsic dimension decreases from regime 1 $\rightarrow$ 2

Conclusion - 2

Why larger models generalize better when trained longer ?

Maybe because:

Stochastic regime acts as concept refinement process

⇒ Manifold dimension reduction ⇒ Better generalization

Future Directions:

- Decomposer-aware intrinsic dimension estimators for structured representations
- Deeper LLM analysis across model sizes, other architectures, and tasks
- Novel concept-based convergence approaches
- Extension to multimodal and continual learning settings

Thank You!

Thank You !

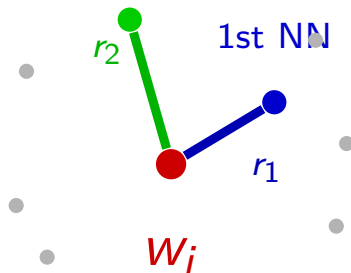
Questions?

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Intrinsic Dimension Estimation: The TWO-NN Theory - 1 [6]

TWO-NN Concept

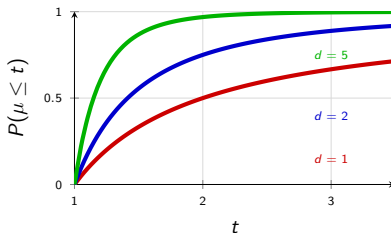
2nd NN



$$\mu_i = \frac{r_2(w_i)}{r_1(w_i)}$$

Core Intuition: The ratio $\mu = r_2/r_1$ follows a theoretical distribution that depends on intrinsic dimension d

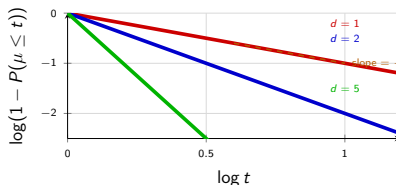
Theory: CDF Relationship



$$P(\mu \leq t) = 1 - t^{-d}$$

Intrinsic Dimension Estimation: The TWO-NN Theory - 2 [6]

Estimation Method: Linear fitting in log-log space



Algorithm Steps:

- 1 For each point w_i , compute distances to 1st and 2nd nearest neighbors
- 2 Calculate ratio $\mu_i = r_2(w_i)/r_1(w_i)$ for all points
- 3 Sort ratios and compute empirical CDF: $P(\mu \leq t)$
- 4 Transform to log-log scale: $\log(1 - P(\mu \leq t))$ vs $\log t$
- 5 Fit linear regression and extract slope $= -\hat{d}$

EuroBERT Training Loss Curve

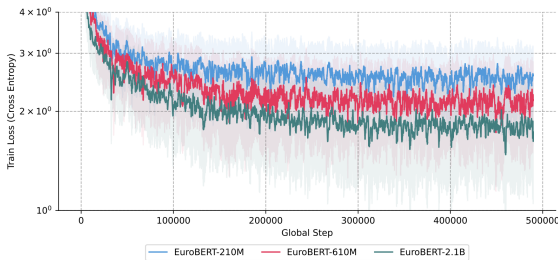


Figure: EuroBERT training loss curve for all family models [4]

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